The empirical recurrence for OEIS A224308, recovered from its tail

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8 September 2026

Abstract

OEIS A224308 records "Empirical recurrence of order 88" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 657$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 11$, so the recurrence is proved for every $n \geq 99$. Its last coefficient is $c_{88} = 95938412544 \neq 0$, so no further tail satisfies anything shorter; any order-88 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A224308 is "Number of nX6 0..2 arrays with diagonals and rows unimodal and antidiagonals nondecreasing.". It has offset 1 and begins

$$148, 4690, 61953, 562760, 4420701, 33934344, 268379906, 2215451575, \ \ldots$$

The entry states:

> Empirical recurrence of order 88 (see link above)

As of the "Last modified" line on the live entry (Oct 03 2025 23:22:04, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 657$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 657$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 88: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 11$ is the first whose minimal order is exactly the stated 88.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1314$ exact terms from $n = 11$ on, it returns order 88 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{88} c_i \, a(n-i),$$

c_{1}	28	c_{23}	1503595330342	c_{45}	−297241926351090938	c_{67}	−2283793762400620032
c_{2}	−283	c_{24}	−4902166987826	c_{46}	−1043821572862888192	c_{68}	−1927869262091505856
c_{3}	1083	c_{25}	8192323587511	c_{47}	−1521567412868618582	c_{69}	−318341168372787712
c_{4}	384	c_{26}	−14724157282344	c_{48}	−2521507835345490336	c_{70}	−1029387659075893760
c_{5}	−7203	c_{27}	23407373455217	c_{49}	−3453314133726720532	c_{71}	355595141047662080
c_{6}	−56172	c_{28}	−88012761010310	c_{50}	−4071989401351047888	c_{72}	59928420380536064
c_{7}	391788	c_{29}	129909757855519	c_{51}	−5048675413563888870	c_{73}	110593948834118656
c_{8}	−556197	c_{30}	−522275750798023	c_{52}	−4288081760518239704	c_{74}	121297300371832832
c_{9}	−670206	c_{31}	494783689183062	c_{53}	−4415549804753178852	c_{75}	−16480397884363776
c_{10}	−2831579	c_{32}	−2279250654893292	c_{54}	−1101459608562796692	c_{76}	11531887864010752
c_{11}	29753831	c_{33}	910190147092487	c_{55}	−91509396397259436	c_{77}	−2616483982047232
c_{12}	−58323686	c_{34}	−2843436202148556	c_{56}	6185677670085966112	c_{78}	−5383115638325248
c_{13}	−2378724	c_{35}	1829684355137015	c_{57}	5538249829044151416	c_{79}	3218797602775040
c_{14}	−114347394	c_{36}	2829184612176290	c_{58}	12935932391354475240	c_{80}	−1226421803483136
c_{15}	1671128038	c_{37}	20761777264718651	c_{59}	7413922607473352968	c_{81}	499447872798720
c_{16}	−4822615816	c_{38}	17110277965939120	c_{60}	12877992313103525808	c_{82}	28822674931712
c_{17}	5240571691	c_{39}	88496785817903180	c_{61}	4343870877847143176	c_{83}	−102940482600960
c_{18}	−7577399804	c_{40}	42383063700039530	c_{62}	8930722104028462640	c_{84}	33618693193728
c_{19}	60166994314	c_{41}	173792858643081965	c_{63}	−175244650097360848	c_{85}	−21413068013568
c_{20}	−216041577377	c_{42}	68769291743358848	c_{64}	5316989732819817792	c_{86}	3584605814784
c_{21}	405752998339	c_{43}	126926977825923122	c_{65}	−2877919989468237376	c_{87}	−676693278720
c_{22}	−642668233113	c_{44}	−128200561241478995	c_{66}	800788957171128512	c_{88}	95938412544

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 99$, and it is the recurrence OEIS A224308 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 11$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{88} - c_1 x^{88-1} - \cdots - c_{88}$. Its constant term is $-c_{88} = -95938412544$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 88 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 88, so $\deg q = 88$, and it is monic. It holds from some index t on. From $\max(t, 11)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 88, so $q = q_{\min}$. ☐

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 88, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 95938412544.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-88 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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